Samia Choueiri

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Samia Choueiri | Samia Choueiri | samiachoueiri

Education

Ph.D. **University of South Carolina**, Informatics — Sep 2023 – Present
- GPA: 4.0/4.0 (Transcript)

MS **American University of Science and Technology**, Computer and Communications Engineering — Oct 2018 – Jun 2021
- Emphasis: Mechatronics Engineering
- GPA: 3.9/4.0 (Degree - Transcript)

BS **American University of Science and Technology**, Computer and Communications Engineering — Oct 2013 – Jun 2018
- Emphasis: Mechatronics Engineering
- GPA: 3.4/4.0 (Degree - Transcript)

Work Experience

Graduate Research/Teaching Assistant, University of South Carolina — Columbia, SC
Department of Integrated Information Technology, Molinaroli College of Engineering and Computing — Sep 2023 – Present

Computer Vision and Application Engineer, Oculi — Rochester, NY
Event-based image sensor development for AI-driven vision applications — Jul 2022 – Jul 2023

Lab Instructor / Teaching Assistant / Senior Project Co-Advisor, American University of Science and Technology — Sep 2018 – Jul 2022
Departments of Mechatronics Engineering, Computer and Communications Engineering, and Biomedical Engineering in the Faculty of Engineering

Academic / Administrative Assistant, American University of Science and Technology — Sep 2018 – Jul 2022
Departments of Mechatronics Engineering, Computer and Communications Engineering, and Biomedical Engineering in the Faculty of Engineering

Robotics and Coding Program Coordinator, Jusoor NGO — Jan 2023 – May 2023

Micro-controller Programming Teacher, College Notre-Dame De Jamhour — Mar 2022 – Mar 2023

Robotics Teacher, École D'Art et De Musique — Apr 2019 – Dec 2021

Technical Scientist, Trawi 3D Concrete Printing Project — Feb 2017 – Jun 2018

Electro-Mechanical Engineer (Internship), Middle East Airlines — Aug 2017 – Oct 2017
Mideast Aircraft Services Company and Organization (MASCO)

Awards

- **Best Paper Award**, IEEE BlackSeaCom 2026, Bucharest, Romania, Jun. 2026.
  Real-Time Detection of Encrypted DGA and DNS Tunneling Using DPDK on SmartNICs.
- **Best Poster Award**, FABRIC KNIT 12 Workshop, Honolulu, HI, Apr. 2026.
  Accelerating Attack and Fault Detection in Cyber-Physical Systems Using SmartNICs and DPDK.
- **Best Poster Runner-up**, USC AI Day, Columbia, SC, Mar. 2026.
  Hardware Offloading for Encrypted Traffic Classification Using P4-DPDK on SmartNICs.
- **Best Poster Award**, FABRIC KNIT 11 Workshop, Salt Lake City, UT, Oct. 2025.

Detection and Mitigation of Volumetric DDoS Attacks using Adaptive Rate Limiting in P4-DPDK.

- **Best Poster Award**, FABRIC KNIT 10 Workshop, Chapel Hill, NC, Mar. 2025.
  Volumetric DDoS Attack Detection and Mitigation using P4-DPDK.
- **Best Poster Runner-up**, FABRIC KNIT 8 & 5NRP Workshop, San Diego, CA, 2024.
  Scalable Heavy Hitter Detection in Virtualized Environments: A DPDK-based Software Approach with P4 Integration.
- **Excellent Oral Presentation Award**, ICARA 2023, Abu Dhabi, United Arab Emirates, 2023.
  Impact Force Location and Intensity Identification Using Joint-Position Sensors in Humanoids.
- **1st Place Winner**, NXP Cup 2021 Lebanon Qualifications, Beirut, Lebanon, 2021.
  Ranking 15th place worldwide.
- **1st Place Winner**, NXP Cup 2019 Lebanon Qualifications, Beirut, Lebanon, 2019.
  Participated in the International Competition at Fraunhofer IIS, Germany.
- **3rd Place Winner**, Engineering Design Competition (EDC 2.0), American University of Beirut, Beirut, Lebanon, 2018.
  Automated Delivery Car, fastest participating car (King of Speed Award).
- **1st Place Winner**, Engineering Expo (Electronics), AUST, Beirut, Lebanon, 2016.
  Rotating Solar Panel.
- **1st Place Winner**, Engineering Expo (Circuits Analysis II), AUST, Beirut, Lebanon, 2016.
  Audio Visualizer.

Publications

[1] L. Albanna, **S. Choueiri**, J. Gomez, E. F. Kfoury, and J. Crichigno, "Live Behavioral Malware Detection through DPU-Based External Memory Introspection," in *2027 IEEE 24th Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, Jan. 2027.

[2] **S. Choueiri**, S. Elizalde, A. Mazloum, A. AlSabeh, J. Gomez, E. F. Kfoury, and J. Crichigno, "Accelerating Attack and Fault Detection in Cyber-Physical Systems Using SmartNICs and DPDK," in *IEEE Global Communications Conference (GLOBECOM)*, Macau, China, Dec. 2026.

[3] A. G. S. P. Narasimha, A. Mazloum, A. G. B. Sai, **S. Choueiri**, E. F. Kfoury, J. Crichigno, and J. Galan-Jimenez, "Offloading Encrypted Traffic Classification to SmartNICs with P4-DPDK," in *International Conference on Network and Service Management (CNSM)*, Madrid, Spain, Oct. 2026.

[4] A. G. B. Sai, A. Mazloum, J. Gomez, **S. Choueiri**, A. G. S. P. Narasimha, E. F. Kfoury, and J. Crichigno, "An Agentic AI Approach for Hands-on Networking Education," in *SIGCOMM Education Workshop 2026: Networking Education for the AI Generation (A4NE)*, Denver, CO, USA, Aug. 2026. DOI: https://a4ne-workshop.github.io/papers/a4ne26-paper5.pdf

[5] S. Elizalde, A. AlSabeh, **S. Choueiri**, J. Galan-Jimenez, E. F. Kfoury, and J. Crichigno, "Real-Time Detection of Encrypted DGA and DNS Tunneling Using DPDK on SmartNICs," in *IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)*, Bucharest, Romania, Jun. 2026. DOI: 10.1109/BlackSeaCom69760.2026.11598333

[6] A. Mazloum, A. G. S. P. Narasimha, **S. Choueiri**, E. F. Kfoury, A. AlSabeh, J. Gomez, and J. Crichigno, "Accelerating DPI Using Domain-Specific Hardware Accelerator for Dynamic Programming," in *IEEE International Conference on Communications (ICC)*, Glasgow, Scotland, UK, May 2026. DOI: 10.1109/ICCWorkshops63917.2026.11586269

[7] A. G. S. P. Narasimha, A. Mazloum, **S. Choueiri**, S. Elizalde, E. F. Kfoury, and J. Crichigno, "Real-Time Encrypted Traffic Classification with P4-DPDK," in *IEEE International Conference on Communications (ICC)*, Glasgow, Scotland, UK, May 2026. DOI: 10.1109/ICC59461.2026.11587847

[8] J. Gomez, **S. Choueiri**, A. Mazloum, S. Elizalde, A. GSPN, A. AlSabeh, E. F. Kfoury, and J. Crichigno, "Design and Deployment of a Testbed for SmartNIC and Programmable Data Plane Experimentation," in *2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC)*, 2026, pp. 1–6. DOI: 10.1109/CCNC65079.2026.11366468

[9] S. Elizalde, **S. Choueiri**, A. Mazloum, J. Gomez, E. Kfoury, and J. Crichigno, "Accelerating Anomaly Detection in Industrial Control Systems Using SmartNICs and DPDK," in *2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC)*, 2026, pp. 1–6. DOI: 10.1109/CCNC65079.2026.11366361

[10] A. Mazloum, S. Elizalde, **S. Choueiri**, E. Kfoury, J. Gomez, A. AlSabeh, and J. Crichigno, "Optimizing Network Resilience Using Domain-Specific Hardware Accelerator for Dynamic Programming," in *Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, New York, NY, USA: Association for Computing Machinery, 2025, pp. 860–865. DOI: 10.1145/3731599.3767449

[11] **S. Choueiri**, A. Mazloum, S. Elizalde, E. F. Kfoury, and J. Crichigno, "Detection and Mitigation of Volumetric DDoS Attacks using Adaptive Rate-Limiting in P4-DPDK," in *GLOBECOM 2025 - 2025 IEEE Global Communications Conference*, 2025, pp. 3329–3334. DOI: 10.1109/GLOBECOM59602.2025.11432232

[12] S. Elizalde, A. AlSabeh, A. Mazloum, **S. Choueiri**, E. Kfoury, J. Gomez, and J. Crichigno, "A survey on security applications with SmartNICs: Taxonomy, implementations, challenges, and future trends," *Journal of Network and Computer Applications*, vol. 242, p. 104 257, 2025. DOI: https://doi.org/10.1016/j.jnca.2025.104257

[13] A. Mazloum, E. Kfoury, **S. Choueiri**, A. AlSabeh, and J. Crichigno, "Offloading Encrypted C2 Traffic Mitigation to SmartNICs," in *2025 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)*, 2025, pp. 1–4. DOI: 10.1109/BlackSeaCom65655.2025.11193948

[14] **S. Choueiri**, A. Mazloum, S. Elizalde, E. F. Kfoury, and J. Crichigno, "Volumetric DDoS Attack Detection and Mitigation using P4-DPDK," in *2025 IEEE International Black Sea Conference on Communications and Networking (BlackSeaCom)*, 2025, pp. 1–6. DOI: 10.1109/BlackSeaCom65655.2025.11193917

[15] S. Elizalde, A. Mazloum, **S. Choueiri**, E. Kfoury, and J. Crichigno, "Performance Evaluation of Stateless Firewalling: Host-Based, SmartNIC, and P4 Switch," in *NOMS 2025-2025 IEEE Network Operations and Management Symposium*, 2025, pp. 1–5. DOI: 10.1109/NOMS57970.2025.11073617

[16] **S. Choueiri**, A. Mazloum, E. F. Kfoury, and J. Crichigno, "Scalable Heavy Hitter Detection: A DPDK-based Software Approach with P4 Integration," in *GLOBECOM 2024 - 2024 IEEE Global Communications Conference*, 2024, pp. 1930–1935. DOI: 10.1109/GLOBECOM52923.2024.10901079

[17] E. F. Kfoury, **S. Choueiri**, A. Mazloum, A. AlSabeh, J. Gomez, and J. Crichigno, "A Comprehensive Survey on SmartNICs: Architectures, Development Models, Applications, and Research Directions," *IEEE Access*, vol. 12, pp. 107 297–107 336, 2024. DOI: 10.1109/ACCESS.2024.3437203

[18] **S. Choueiri**, H. Diab, M. Owayjan, and R. Achkar, "Impact Force Location and Intensity Identification Using Joint-Position Sensors in Humanoids," in *2023 9th International Conference on Automation, Robotics and Applications (ICARA)*, 2023, pp. 112–116. DOI: 10.1109/ICARA56516.2023.10125728

[19] **S. Choueiri**, D. Daoud, S. Harb, and R. Achkar, "Fire and Smoke Detection Using Artificial Neural Networks," in *2020 14th International Conference on Open Source Systems and Technologies (ICOSST)*, 2020, pp. 1–6. DOI: 10.1109/ICOSST51357.2020.9332990

[20] **S. Choueiri**, M. Owayjan, H. Diab, and R. Achkar, "Mobile Robot Path Planning Using Genetic Algorithm in a Static Environment," in *2019 Fourth International Conference on Advances in Computational Tools for Engineering Applications (ACTEA)*, 2019, pp. 1–6. DOI: 10.1109/ACTEA.2019.8851100

Grant Participation

Contributed to proposal writing and served as lead developer under the direction of the Principal Investigator on the following funded projects at the University of South Carolina.

- Cyber-Con2: Multi-sector Convergence to Advance the Preparation of Learners for OT and IT Cybersecurity Convergence Workforce *Sep 2024 – Aug 2027*
  NSF DUE #2400729 | $650,000 | PI: Jorge Crichigno
  Contribution: Designed a system to detect faults and attacks in cyber-physical systems; main developer of the Zeek IDS lab series.

- CyberTraining on Accelerating Infrastructure Workloads using Next-Generation SmartNICs/DPUs *Aug 2024 – Jul 2027*
  NSF OAC #2417823 | $790,000 | PI: Elie Kfoury
  Contribution: Main developer of lab libraries introducing P4-DPDK, security applications, and Zeek programming; assisted in developing SmartNIC programming labs using NVIDIA DOCA and P4-DOCA.

- Enhancing Network Security by Implementing an ML Malware Detection and Classification Scheme in P4 Programmable Data Planes and SmartNICs *Jul 2024 – Jun 2027*
  NSF OAC #2403360 | $599,999 | PI: Jorge Crichigno
  Contribution: Developed data-plane solutions for blocking malicious traffic and mitigating heavy-flow attacks.

- Enhancing Data Transfers by Enabling Programmability and Closed-loop Control in a Non-programmable Science DMZ — *Jul 2024 – Jun 2026*
  NSF OAC #2346726 | $500,000 | PI: Jorge Crichigno
  Contribution: Primary funding for doctoral research: developed P4-DPDK schemes for mitigating DDoS attacks in the data plane; published the work and presented it at workshops and student presentation sessions.

- Preparing Cyber Warfare Professionals by Integration of Curriculum, Experiences, and Internships — *Feb 2023 – Jan 2026*
  ONR #N00014-23-1-2245 | $600,000 | PI: Jorge Crichigno
  Contribution: Mentored undergraduate students on cybersecurity projects using next-generation firewalls.

- CyberTraining on P4 Programmable Devices using an Online Scalable Platform with Physical and Virtual Switches and Real Protocol Stacks — *Oct 2021 – Sep 2025*
  NSF OAC #2118311 | $499,540 | PI: Jorge Crichigno
  Contribution: Authored a comprehensive survey on SmartNICs.

- Enhancing the Preparation of Next-generation Cyber Professionals through a Hands-on Academic Program, Undergraduate Research Experiences, and Community Partnerships — *Jul 2020 – May 2022*
  ONR #N00014-20-1-2797 | $249,997 | PI: Jorge Crichigno
  Contribution: Supervised undergraduate cybersecurity projects using Palo Alto NGFWs and P4, including stateful packet filtering, policy-based forwarding, geoblocking, App-ID, and DDoS detection and prevention.

Teaching Experience

University of South Carolina, Columbia, SC Evaluations — **2025 – Present**

Introduction to Computer Hardware and Software | ITEC 233 — *Fall 2025, Fall 2026*
Course on identifying, installing, configuring, and troubleshooting PC components, operating systems, mobile devices, and networks. Topics include PC assembly, preventive maintenance, networking, Windows installation and configuration, mobile and alternative operating systems, virtualization, cloud computing, security, and the role of the IT professional. Delivered through the Cisco Networking Academy IT Essentials platform.

Thriving in the Tech Age | ITEC 101 — *Spring 2026*
Broad introduction to information technology and the computing disciplines, IT careers, industry certifications, and foundational computing skills. Examines the ethical, social, and global implications of technology, professional values and social responsibility (VESR), and the impact of emerging, disruptive, and converging technologies, while developing technical communication, teamwork, and critical-thinking skills.

American University of Science and Technology, Beirut, Lebanon Evaluations — **2019 – 2022**

Senior Design Project Advisor | CCE/MTE/BME 497X & 499X — *Fall 2021, Spring 2022*
Advised senior capstone teams in Computer and Communications, Mechatronics, and Biomedical Engineering through the two-course sequence: Senior Design Proposal (497X) and Senior Design (499X).

Computer Organization & Microprocessors | CCE 320, CCE 320L — *Fall 2021, Spring 2022*
Lecture (Spring 2022) and laboratory (2 terms). The lab provides hands-on experiments and projects: Arduino hardware and programming, interfacing exercises, and programming and control of external tasks.

Mechanics of Materials | MTE 310, MTE 310P — *Spring 2021 – Fall 2021*
Lecture (Summer 2021) and supplementary problem-solving sessions (3 terms): stress and strain, axial loading, torsion, bending, transverse shear, combined loading, stress transformation, and beam deflection.

Problem Solving & Circuit Analysis II | CCE 202P — *Spring 2021*
Supplementary problem-solving sessions for CCE 202: phasors and sinusoidal analysis, frequency response, power calculations, three-phase circuits, Laplace and Fourier methods, Bode diagrams, and active filters.

Circuits I Laboratory | CCE 201L — *Fall 2020 – Spring 2022*
Laboratory instruments and devices, electrical materials, and soldering techniques; simple electric and electronic circuit applications; PCB design and implementation; measurements and transformers (5 terms).

Digital Systems Laboratory | CCE 220L — *Fall 2020 – Spring 2022*
Logic gates and properties of TTL and CMOS families; combinational design with decoders, multiplexers, ROMs, and displays; flip-flops and synchronous sequential circuits; counters, adders, and ALU design (5 terms).

Problem Solving in Electronics | CCE 301P — *Fall 2020 – Spring 2022*

Supplementary problem-solving sessions for CCE 301 (5 terms): semiconductor theory, PN junctions, ideal op-amps, BJTs, FETs, DC biasing, I–V characteristics, single-stage amplifiers, low-frequency small-signal models, oscillators, filters, power supplies, and voltage regulation.

Problem Solving in Dynamics | MTE 212P *Fall 2020 – Spring 2021*
Vector description of force, position, velocity, and acceleration in fixed and moving reference frames; kinetics of particles, systems of particles, and rigid bodies; energy and momentum methods; Euler's equations; and moments of inertia.

Problem Solving in Solid Mechanics | MTE 211P *Fall 2019*
Supplementary problem-solving sessions for MTE 211, covering the theory and application of the principles of engineering mechanics to rigid bodies.

Teaching Materials Developed

NSF-funded lab series combining step-by-step manuals with virtual machines, freely available on NETLAB, the NSF FABRIC testbed, and GitHub, and used in the workshops listed in this CV.

- Introduction to Intrusion Detection Using Zeek *6 labs*
  Lead developer | Deployed on NETLAB
  Zeek scripting, events, and custom logging; detecting ARP spoofing and DNS amplification attacks.
- DPU Programming using P4 *7 labs*
  Contributing developer | Deployed on NETLAB
  P4 and DOCA Pipeline Language (DPL), custom parsers, packet mirroring, match-action tables, P4Runtime, and counters.
- DPU Programming using DOCA *5 labs*
  Contributing developer | Deployed on NETLAB
  DOCA environment setup, device and memory management, the progress engine, and hardware accelerator offloading.
- Introduction to P4-DPDK on FABRIC *9 labs*
  Lead developer | Deployed on FABRIC testbed FABRIC artifact GitHub
  Creating, testing, and deploying P4-DPDK applications on custom FABRIC topologies, including multicore pipeline processing.
- P4-DPDK Security Lab Series *8 labs*
  Lead developer | Deployed on NETLAB
  Packet interarrival times with hashes and registers, SYN flood mitigation, heavy-hitter detection with count-min sketches, and stateful packet filtering.
- Introduction to P4-DPDK *7 labs*
  Lead developer | Deployed on NETLAB
  P4 building blocks with the PNA architecture, parser implementation, match-action tables, runtime table management, and deparsing.

Workshops & Training Delivered

- Hands-on Workshop on Science DMZ, Zeek IDS, and P4-DPDK *May 2026*
  CyberTraining Workshop 2026, Online | 176 attendees
- DPU Programming using DOCA *Apr 2026*
  FABRIC KNIT 12, University of Hawaii at Mānoa, Honolulu, HI | 25 attendees
- Hands-on Workshop on Science DMZ and P4-DPDK *Aug 2025*
  CyberTraining Workshop Summer 2025, Online | 103 attendees
- Cybersecurity and P4-DPDK Workshop *May 2025*
  Minority Serving – Cyberinfrastructure Consortium Annual Meeting, Nashville, TN | 22 attendees
- Hands-on Workshop on Science DMZ and P4-DPDK *Apr 2025*
  CyberTraining Workshop Spring 2025, Online | 117 attendees

- Developing High Performance Networked Applications using P4-DPDK *Mar 2025*
  FABRIC KNIT 10, University of North Carolina, Chapel Hill, NC | 20 attendees
- Developing Defenses with P4 and DPDK *Oct 2024*
  NSF Cybersecurity Summit 2024, Carnegie Mellon University, Pittsburgh, PA | 91 attendees
- Hands-on Workshop on Science DMZ and P4-DPDK *Aug 2024*
  CyberTraining Workshop 2024, Online | 146 attendees
- Python Game Development: Building Five Games and a Final Coding Competition *May 2023*
  Python Coding Workshops, American University of Science and Technology, Beirut, Lebanon | 250 students
- Building Arduino Soccer Robots and a Final Robot Soccer Competition *Jul 2019*
  AUST Arduino Summer Camp, American University of Science and Technology, Beirut, Lebanon | 150 students
- Building Line-Following Racing Robots and a Final Race Competition *Jul 2018*
  AUST Arduino Summer Camp, American University of Science and Technology, Beirut, Lebanon | 100 students

Invited & Refereed Talks

- Accelerating Attack and Fault Detection in Cyber-Physical Systems Using SmartNICs and DPDK *Apr 2026*
  FABRIC KNIT 12, Honolulu, HI
- Detection and Mitigation of Volumetric DDoS Attacks using Adaptive Rate Limiting in P4-DPDK *Oct 2025*
  FABRIC KNIT 11, Salt Lake City, UT
- Building a Career in Cybersecurity Research *Jul 2025*
  IEEE Women in Engineering Day 2025, American University of Science and Technology, Beirut, Lebanon
- Real-Time Encrypted Traffic Classification with P4-DPDK *Mar 2025*
  FABRIC KNIT 10, Chapel Hill, NC

Poster Presentations

- An Agentic AI Approach for Hands-on Networking Education *Sep 2026*
  FABRIC KNIT 13, Washington, DC
- Real-Time Detection of Encrypted DGA and DNS Tunneling Using DPDK on SmartNICs *Apr 2026*
  FABRIC KNIT 12, Honolulu, HI
- Accelerating Attack and Fault Detection in Cyber-Physical Systems Using SmartNICs and DPDK *Apr 2026*
  FABRIC KNIT 12, Honolulu, HI
- Runtime Memory Introspection for Behavioral System Monitoring Using DPUs *Apr 2026*
  FABRIC KNIT 12, Honolulu, HI
- Detection and Mitigation of Volumetric DDoS Attacks using Adaptive Rate Limiting in P4-DPDK *Mar 2026*
  USC AI Day, Columbia, SC
- Real-Time Detection of Encrypted DGA and DNS Tunneling Using DPDK on SmartNICs *Mar 2026*
  USC AI Day, Columbia, SC
- Hardware Offloading for Encrypted Traffic Classification Using P4-DPDK on SmartNICs *Mar 2026*
  USC AI Day, Columbia, SC
- Detection and Mitigation of Volumetric DDoS Attacks using Adaptive Rate Limiting in P4-DPDK *Oct 2025*
  FABRIC KNIT 11, Salt Lake City, UT
- Volumetric DDoS Attack Detection and Mitigation using P4-DPDK *May 2025*
  Minority Serving – Cyberinfrastructure Consortium Annual Meeting, Nashville, TN
- Volumetric DDoS Attack Detection and Mitigation using P4-DPDK *Mar 2025*
  FABRIC KNIT 10, Chapel Hill, NC

- Real-Time Encrypted Traffic Classification with P4-DPDK *Mar 2025*
  FABRIC KNIT 10, Chapel Hill, NC
- Offloading Encrypted C2 Traffic Mitigation to DPDK *Mar 2025*
  FABRIC KNIT 10, Chapel Hill, NC
- Scalable Heavy Hitter Detection in Virtualized Environments: A DPDK Approach with P4 Integration *Aug 2024*
  TechNet Augusta, Augusta, GA
- Scalable Heavy Hitter Detection in Virtualized Environments: A DPDK Approach with P4 Integration *Mar 2024*
  FABRIC KNIT 8 & 5NRP, San Diego, CA

Technical Skills

Networking & Security	Programmable data planes (P4), kernel-bypass packet processing (DPDK), next-generation firewalls (Palo Alto), intrusion detection (Zeek, Suricata), DDoS detection and mitigation, IoT, industrial networking (PROFINET)
Applied AI/ML for Security	ML-based network traffic classification, encrypted and malicious traffic detection, anomaly and attack detection, offloading ML-based detection to SmartNICs and the data plane
Programming & Scripting	P4, C, C++, Python, Zeek scripting, shell scripting, multi-threading
Frameworks & SDKs	DPDK, P4-DPDK, NVIDIA DOCA, P4Runtime
Hardware Platforms	Intel Tofino P4 switches, SmartNICs/DPUs (NVIDIA BlueField, Intel IPU), enterprise and HPC rack servers (Dell PowerEdge, Lenovo ThinkSystem)
Cloud & Virtualization	Proxmox VE, VMware (ESXi, vSphere), AWS; testbeds: FABRIC, NETLAB
Embedded & Engineering Tools	Arduino, Raspberry Pi, NI myDAQ/myRIO, PLC, PIC, PCB design, 3D printing, MATLAB, LabVIEW, Multisim, ROS, Gazebo, Autodesk Inventor, SolidWorks

Data Center Operations

- **Academic cloud:** administered the virtualized infrastructure hosting the NETLAB virtual labs, including VM provisioning, resource allocation, and user access for courses and national workshops.
- **Rack operations:** managed two equipment racks, covering hardware installation and integration of compute and network equipment, with centralized monitoring and management through Lenovo XClarity and Dell OpenManage.
- **Server provisioning:** deployed bare-metal servers and hypervisor hosts, including OS installation and network configuration.
- **Programmable hardware:** installed, configured, and integrated Intel Tofino P4 switches and SmartNICs/DPUs (NVIDIA BlueField, Intel IPU).
- **Research testbeds:** designed and deployed the network topologies used in research experiments.

Certifications

Industrial Networking & Cybersecurity

- Industrial Cybersecurity Essentials *Aug 2026*
  Cisco Networking Academy, Stanly Community College
- Industrial Networking Essentials *Aug 2026*
  Cisco Networking Academy, Stanly Community College
- Industrial IoT and Control Systems in Manufacturing *Aug 2026*
  Cisco Networking Academy, Stanly Community College
- Certified PROFINET Engineer, PROFINET Installer, and PROFINET PA Engineer *Dec 2025*
  PROFIBUS & PROFINET International (PI), Johnson City, TN

Networking & Security

- CCNA: Enterprise Networking, Security, and Automation *Nov 2026*
  Cisco Networking Academy, Stanly Community College

- CCNA: Switching, Routing, and Wireless Essentials — *May 2026*
  Cisco Networking Academy, Stanly Community College
- CyberOps Associate — *Jul 2025*
  Cisco Networking Academy, Stanly Community College
- Cisco Certified Instructor, CCNA: Introduction to Networks — *May 2025*
  Cisco Networking Academy, instructor training course
- Proxmox VE 8: Setup and Manage — *Apr 2025*
  Stanly Community College
- Introduction to DOCA for DPUs — *Dec 2023*
  NVIDIA Deep Learning Institute

Professional & Other

- Certified Peer Reviewer — *Feb 2026*
  ACM Publications Board, Peer Reviewer Training Course
- Robot Operating System (ROS) Training, DECAIR — *Dec 2021*
  University of Pisa
- SolidWorks Workshop, Certificate of Participation — *Dec 2017*
  Berytech Fab Lab, Mar Roukoz, Lebanon
- International Certification of Digital Literacy (ICDL), all seven modules — *May 2009*
  ICDL Foundation

Professional Affiliations

- IEEE Student Member
- IEEE Communications Society (ComSoc)
- IEEE Women in Engineering (WIE)

Professional Service

Peer Review

- Journal Reviewer — *2026 – present*
  IEEE Networking Letters (IEEE Xplore); Scientific Reports (Nature); CMES (Tech Science Press)
- Conference Reviewer — *2026 – present*
  ACDSA 2027; ICECER 2026; ICECET 2026

Judging

- Judge, PalmettoHacks 2026 — *Oct 2026*
  24-hour MLH hackathon hosted by Kappa Theta Pi, University of South Carolina, Columbia, SC
- Judge, VEX Robotics Competition — *Dec 2023*
  Middle and high school competition with about 50 teams, hosted by the Gamecock Robotics Club, University of South Carolina, Columbia, SC

Languages

Arabic	Native
English	Full professional proficiency | IELTS Academic 7.5 (CEFR C1), Jun. 2023
French	Working knowledge